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Market Structure in the Age of AI: From Software Scale to Physical Constraint

Jordi Visser

JVisser@22VResearch.com

22vresearch.com

Over the last 17 years since the Global Financial Crisis, markets have not simply experienced cycles, they have been structurally rewired. Post-GFC policy responses combined with exponential technological innovation reshaped how capital flows, how risk is managed, and how prices are formed. What emerged was a system increasingly driven by reflexive flows, mechanical allocation, and dominating price momentum.

On the asset-management side, exponential innovation accelerated a historic migration from active to passive investing. As software and platform businesses scaled globally with minimal marginal cost and low debt, identifying sustainable alpha through traditional stock selection became more difficult. Passive vehicles, allocating capital mechanically by market capitalization, rose to dominance, reinforcing price trends as winning companies automatically attracted more capital. Even active mutual funds, tethered to these indices for benchmarking, were compelled to minimize tracking error by crowding into the same mega-cap leaders, effectively amplifying the passive flow. The concentration extended past market cap and into performance. Since December 31st, 2019 the S&P 1500 is up 106% and only 27% of the stocks in the index have outperformed that return highlighting the concentration of returns. Markets gradually shifted from valuation-driven pricing toward flow-driven reinforcement.

Hedge fund allocations adapted by moving away from high-beta, concentrated, net-long exposure toward factor-neutral, multi-strategy, and quantitative approaches. Regulatory constraints, volatility suppression, and increased crowding reduced the effectiveness of discretionary risk-taking. Instead, funds arbitrated correlations, factors, and volatility itself. While this created surface-level stability, it also increased systemic fragility, as many strategies now respond to the same signals and de-risk simultaneously during periods of stress.

Retail participation further altered market structure, particularly through the rise of options trading and zero-days-to-expiration contracts. Zero-commission platforms transformed retail flow into a structural driver of short-term price action, but more importantly revealed a distinct behavioral pattern: impatience combined with recency bias. Retail investors tend to gravitate toward established winners, using options as leverage to accelerate gains in assets already in motion. This is not portfolio construction, it is momentum-chasing, an instinct to make trends move faster.

As a result, retail flow becomes highly concentrated in the same momentum leaders already benefiting from passive allocations and systematic strategies, creating a third reinforcing layer of demand. Dealer hedging of these concentrated options positions further amplifies price movements, producing volatility driven by positioning, gamma dynamics, and reflexive flows rather than changes in fundamentals. When momentum ultimately breaks, retail investors are often the last to exit, but their concentrated exposure in former leaders accelerates the reversal, turning amplification into liquidation.

Exponential innovation, most recently through artificial intelligence, also drove extreme concentration in market leadership, elevating momentum from a factor to the system's organizing principle. AI-enabled platforms and hyperscalers benefited from winner-take-most economics, allowing a narrow group of companies to dominate index returns. Passive inflows, systematic strategies, options-related flows, and historically large corporate buybacks reinforced this momentum, extending trends far longer than traditional models would have anticipated. Investors should remember that momentum is a chameleon factor. During macro regime shifts, today's winners can quickly become tomorrow's losers because positioning is built around relative earnings revisions from the prior environment. This dynamic has historically appeared

around recessions, but it also emerges when PMIs inflect, moving from contraction to expansion or from expansion to slowdown. In these transitions, it is not the passive bid that drives reversals, but active managers and systematic strategies simultaneously repositioning as earnings expectations shift, turning former momentum leaders into the primary fuel for mean reversion.

Given the historic concentration of capital, the market is uniquely vulnerable to a shift in leadership, but this would not be a typical post GFC cyclical rotation, such as the PMI rebound following the oil collapse in 2015 or the post-COVID reopening surge. Instead, the software-over-hardware regime that has dominated for the past 15 years is colliding with a structural transition in how growth is achieved. AI is entering a phase where acceleration is constrained not by code, but by physical scarcity. Continued progress now depends on massive data-center buildouts, insatiable power demand, specialized hardware, and the integration of intelligence into machines operating in the physical world. At the same time, the democratization of code and AI-driven productivity gains threaten to compress the revenue-per-employee advantages that once justified premium valuations for software-centric businesses. As a result, market capitalizations increasingly risk reflecting backward-looking assumptions in a world where competitive moats are shifting toward underinvested infrastructure, energy access, and execution rather than software alone.

This transition also weakens one of the most powerful sources of self-reinforcing equity demand: buybacks. Hyperscalers that once returned excess free cash flow to shareholders are now redirecting capital toward unprecedented AI-related capex, increasingly funded by debt. At the same time, large private growth companies such as xAI, Anthropic, and SpaceX face similar capital intensity, making public markets a likely funding destination. The result may be a wave of major IPOs over the next 12 to 18 months, increasing equity supply just as buyback demand fades.

Taken together, these forces suggest the market is approaching an inflection point rather than an endpoint. The post-GFC regime rewarded software-driven scalability, passive flows, buybacks, and momentum reinforced by abundant liquidity and low physical constraints. That regime is now colliding with a world where growth increasingly requires real-world inputs, power, chips, factories, data centers, and embodied intelligence, while equity supply rises and self-reinforcing demand weakens. Markets built on backward-looking assumptions of capital-light expansion may need to reprice toward capital intensity, infrastructure scarcity, and execution risk. The next phase may not be defined by slower innovation, but by a reordering of value as intelligence moves from code into the physical world.

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